

Hereditary Combined Deficiency of the Vitamin K-Dependent Coagulation Factors

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Abstract

Hereditary combined vitamin K-dependent coagulation factor deficiency (VKCFD) is an extremely rare autosomal recessive genetic disorder characterized by deficiencies in vitamin K-dependent coagulation factors and natural anticoagulants. The condition presents with a spectrum of bleeding symptoms ranging from mild to severe, often beginning in the neonatal period. These bleeding episodes can be particularly severe and even life-threatening, occurring spontaneously or during surgery. In addition to bleeding problems, individuals with VKCFD may experience a variety of non-hemostatic problems, including skeletal deformities, cardiovascular abnormalities, and skin conditions.

VKCFD is caused by variants in the genes encoding either γ -glutamyl carboxylase or the vitamin K 2,3-epoxide reductase complex. Both proteins play a critical role in γ -carboxylation, a posttranslational modification that is essential for the proper function of vitamin K-dependent proteins. Timely and accurate diagnosis is essential to differentiate VKCFD from other genetic and acquired disorders, and genetic testing is required to identify the specific variant.

The primary treatment for VKCFD is the administration of vitamin K, with transfusions of fresh frozen plasma often required during surgery or in cases of severe bleeding. In certain situations, alternative therapies such as prothrombin complex concentrates (PCCs) or a combination of recombinant activated factor VII and vitamin K may be considered. With appropriate treatment, individuals with VKCFD generally have a good clinical outcome, and the condition has a limited impact on their quality of life. This article presents a comprehensive review of all 57 VKCFD cases documented in the literature, as well as 4 new, unpublished cases from France.

Keywords

- VKCFD
- *GGCX*
- *VKORC1*
- bleeding
- vitamin K-dependent coagulation factors

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Introduction

Hereditary combined vitamin K-dependent coagulation factor deficiency (VKCFD) is an extremely rare autosomal recessive disorder characterized by a deficiency of multiple vitamin K-dependent coagulation factors. It results from genetic defects affecting more than one coagulation factor and is inherited as a familial trait. VKCFD can result from variants in either γ -glutamyl carboxylase (*GGCX*) or the vitamin K 2,3-epoxide reductase complex subunit 1 (*VKORC1*), both of which play a critical role in the vitamin K cycle.¹

This article aims to provide an update on VKCFD, followed by a comprehensive review of all published cases, including new, previously unpublished cases from France.

Essential Insights into Vitamin K

Vitamin K consists of lipophilic molecules with a 2-methyl-1,4-naphthoquinone core and a polyisoprenoid side chain. The length and saturation of this side chain classify the different forms of vitamin K.² There are three main forms: (i) Vitamin K1 is mainly found in green leafy vegetables (e.g., spinach, cabbage, broccoli) and vegetable oils, providing 75 to 90% of dietary vitamin K intake in humans. It is also commonly used in supplements and medications. (ii) Vitamin K2 is produced by gut bacteria and found in animal products (e.g., liver, fish, eggs) and fermented foods (e.g., cheese, soy). (iii) Vitamin K3 (menadione) is a synthetic, water-soluble form.^{2–4}

Like other fat-soluble vitamins, dietary vitamin K is incorporated into mixed micelles in the intestinal lumen, which include bile salts, phospholipids, triglycerides, cholesterol, and other lipids. Absorption occurs in the jejunum through active transport. Once absorbed, vitamin K is packaged into chylomicrons and secreted into the mesenteric lymphatic system, entering the bloodstream via the thoracic duct. Vitamin K is also synthesized endogenously.

Vitamin K acts as a cofactor for *GGCX*, an enzyme found throughout the body, with high concentrations in the liver and bone. *GGCX* catalyzes the posttranslational carboxylation of glutamic acid (Glu) residues to γ -carboxyglutamic acid (Gla) on vitamin K-dependent proteins, a modification crucial for their biological activity.^{5,6} During this process, dietary vitamin K quinone is converted into vitamin K epoxide ($VK > O$), an oxidized form. As part of the vitamin K regeneration cycle, $VK > O$ is reduced back to vitamin K quinone and then to vitamin K hydroquinone (VKH_2) through a two-step process involving the enzyme *VKORC1*.⁶ *GGCX* also oxidizes VKH_2 to $VK > O$. The formation of Gla residues allows these proteins to bind divalent ions, particularly calcium (Ca^{2+}) and magnesium (Mg^{2+}), which are essential for their function.⁵

Vitamins K1 and K2 share a common catabolic pathway in hepatocytes. Their side chains undergo ω -hydroxylation by Cytochrome P450 F2 (*CYP4F2*) in the endoplasmic reticulum, followed by side-chain shortening through β -oxidation by mitochondrial trifunctional protein. The metabolites are then conjugated by UDP-glucuronyl transferase. Vitamin K metabolites are primarily excreted via bile (40–60%) and urine (20%).^{2,3}

Vitamin K-Dependent Proteins: Essential Players in Calcium Binding and Physiological Processes

Vitamin K-dependent proteins are a specialized group of proteins that undergo posttranslational modification by vitamin K-mediated γ -carboxylation of specific glutamic acid (Glu) residues, converting them to γ -carboxyglutamic acid (Gla) residues. This modification is critical for their biological activity, particularly their ability to bind calcium ions (Ca^{2+}), which is essential for their functional roles.^{6,7} In cases of vitamin K deficiency or inhibition by antagonists such as 4-hydroxycoumarins, which block *VKORC1*, non- or undercarboxylated proteins accumulate. These are called Proteins Induced by Vitamin K Absence or Antagonism (PIVKA).⁸

The γ -carboxylation process generates Gla residues, which have a high affinity for calcium ions. This property is critical for the activity of vitamin K-dependent proteins, particularly those involved in blood clotting, bone mineralization, and vascular function.

Key vitamin K-dependent proteins include coagulation factors II (prothrombin), VII, IX, and X, which are essential for blood clotting, and proteins C, S, and Z, which regulate coagulation processes.⁹ Certain bone matrix proteins, such as osteocalcin—critical for bone mineralization and skeletal health—and matrix Gla protein (MGP), expressed in the connective tissue, nephrocalcin A–D, Gas 6 protein, transmembrane Gla proteins 3 and 4, are also vitamin K-dependent.⁹ Most non-hemostatic manifestations observed in individuals with VKCFD stem from the defective synthesis of key vitamin K-dependent proteins. These proteins are crucial for maintaining balance in critical systems such as cardiovascular and skeletal health. The γ -carboxylation process enables their functionality by facilitating interactions with calcium and other molecules, which are vital for supporting essential physiological processes.

Hereditary Deficiency of Vitamin K-Dependent Coagulation Factors: A Rare and Complex Bleeding Disorder

Hereditary deficiency of vitamin K-dependent coagulation factors (VKCFDs) is an extremely rare autosomal recessive bleeding disorder caused by defective γ -carboxylation of coagulation factors II (prothrombin), VII, IX, and X, and anticoagulant proteins C, S, and Z. This defect impairs the activity of these key proteins, resulting in a predisposition to bleeding due to dysfunction in coagulation and regulatory pathways.¹⁰ VKCFD results from genetic variants affecting enzymes in the vitamin K cycle and is classified into two subtypes.

Type 1 (VKCFD 1): This is caused by variants in the *GGCX* gene located on chromosome 2 (2p11.2). *GGCX* is critical for the γ -carboxylation process, and complete loss of its activity is likely embryonically lethal, as shown by knockout mouse models. Most cases of VKCFD 1 are due to missense variants in the coding regions of the *GGCX* gene, although there are exceptions, such as homozygous deletions (e.g., c.1056_1059del in intron 1) and splice site abnormalities.¹¹

Type 2 (VKCFD 2): This is caused by variants in the *VKORC1* gene located on chromosome 16 (16p11.2). One known variant in this gene is a homozygous missense variant (c.292C>T) found in three unrelated families. This variant disrupts the retention signal of *VKORC1* in the endoplasmic reticulum and impairs its function.¹

These genetic findings highlight the heterogeneity of VKCFD variants and their diverse effects on the vitamin K cycle. The rarity and genetic complexity of the disorder underscore the need for detailed genetic and functional studies to ensure accurate diagnosis and treatment. The first case of VKCFD was reported in 1966 in an infant girl with a history of bleeding since birth. The proband had significantly low or undetectable levels of coagulation factors II, VII, IX, and X without evidence of liver disease or malabsorption.¹² To date, about 30 families have been documented across multiple continents, including the Americas, Europe, North Africa, the Middle East, and Asia, with no specific ethnic predisposition.⁹ Consanguinity is common among affected families. Homozygous variants occur in more than half of the documented cases.

Diagnosis and Differential Diagnosis

Clinical Manifestations

Patients with VKCFD can present with both hemostatic and non-hemostatic clinical manifestations. The bleeding tendency in these patients varies significantly and is primarily influenced by the plasma levels of vitamin K-dependent proteins. Severe deficiencies in coagulation factors can lead to early hemorrhagic symptoms at birth, such as bleeding at cord separation, intracranial hemorrhage, or spontaneous joint bleeding.^{13–16} Moderate deficiencies often result in cutaneous and mucosal bleeding, prolonged bleeding after invasive procedures,^{17–20} and spontaneous gastrointestinal bleeding, particularly during antibiotic or anticonvulsant treatments, which can disrupt intestinal flora and lead to vitamin K deficiency. In some cases, patients with mild deficiencies may remain asymptomatic.

Defective carboxylation of non-hemostatic proteins, particularly MGP and Gla-rich protein (GRP), may contribute to additional non-hemostatic symptoms, although these are not consistently observed. For example, patients may present with skeletal abnormalities that resemble those observed in warfarin embryopathy. These striking non-hemostatic features include stippling of the long bone epiphyses and shortening of the distal phalanges of the fingers.²¹ Additionally, osteoporosis may occur, often without the typical circulating serum markers of bone remodeling. Cardiac abnormalities, including septal defects and pulmonary artery stenosis, may also be observed. This phenotype is similar to Keutel syndrome, which is associated with MGP variants. It is characterized by diffuse cartilage calcification, brachytelephalangism, peripheral pulmonary artery stenoses, and facial dysmorphism.²²

Some patients may also have elastic fiber dystrophy, characterized by loss of dermal elasticity (resulting in a cutis laxa appearance), subretinal Bruch's membrane

changes (e.g., angioid streaks, "peau d'orange" appearance), and early-onset atherosclerosis.^{23,24} Similar features are observed in pseudoxanthoma elasticum (PXE), a condition associated with *ABCC6* variants. PXE is associated with ectopic calcification due to reduced levels of inorganic pyrophosphate, a key inhibitor of calcification. Notably, two consanguineous families with a PXE-like syndrome have been reported. In these families, 14 affected individuals carried a homozygous splice site variant (c.373+3G>T) in the *GGCX* gene. These patients exhibited PXE-like symptoms without coagulation factor deficiencies, suggesting an alternative pathophysiological mechanism distinct from classic coagulation-related disorders.^{25,26}

Laboratory Assessment

Laboratory testing is critical to the diagnosis of VKCFD. The disease should be suspected in patients with a prolonged prothrombin time (PT), with or without a prolonged activated partial thromboplastin time (aPTT), especially in the presence of hemorrhagic symptoms, although these symptoms are not always present. VKCFD is characterized by prolonged PT and, in some cases, prolonged aPTT.⁹ The degree of prolongation depends on the reduction of specific coagulation factors, with factor VII (FVII) being the most sensitive due to its short half-life.²⁷ Consequently, PT is usually more prolonged than aPTT. Importantly, both PT and aPTT normalize when mixed with normal plasma, indicating deficiency rather than inhibition. Vitamin K-dependent coagulation factors (FII, FVII, FIX, and FX) and anticoagulant proteins (C, S, and Z) show variable reductions in activity, ranging from 1% to 70% in patients. The risk of bleeding is influenced by the severity of these deficiencies, the patient's bleeding history, and the nature of any planned procedures. Individualized management strategies are essential to minimize the risk of bleeding complications.

Genotyping plays a critical role in the accurate diagnosis of VKCFD. Genetic analysis of the *GGCX* gene (spanning 13 kb and containing 15 exons) and the *VKORC1* gene (spanning 5 kb and containing 3 exons) provides a definitive diagnosis, distinguishing VKCFD from other causes of coagulation abnormalities. This precise identification is essential for guiding patient management and family counseling.^{9,28}

While genotyping is the gold standard, additional biomarkers can offer supportive diagnostic information, although they are not routinely used in clinical practice. These include (i) plasma vitamin K1 levels, measured by high-performance liquid chromatography (HPLC), where a serum level below 0.15 µg/L in non-fasting individuals suggests deficiency; (ii) VK > O levels, which indicate disruptions in the vitamin K cycle; (iii) non-carboxylated Factor II (PIVKA-II) levels, measured by ELISA, which are highly sensitive but lack specificity. PIVKA-II cannot differentiate between hereditary VKCFD and acquired vitamin K deficiency caused by dietary insufficiency, intestinal malabsorption syndromes, liver disease, rodenticide poisoning, or drug-induced vitamin K metabolism disorders (→ Table 1).

Table 1 Drugs responsible for vitamin K deficiency

Drug class	Molecule(s)	Mechanism
VKA	Coumarins and derivatives	Inhibition of VKORC1 activity; decrease of reduced vitamin K available for GGCX
Antiepileptic drugs	Phenytoin Carbamazepine	Interaction with liver CYP450 leads to increased vitamin K catabolism
Antituberculous drugs	Rifampicin Isoniazid	Interaction with liver CYP450 leads to increased vitamin K catabolism
Antibiotics	Broad-spectrum antibiotics, such as ampicillin, ceftriaxone, imipenem, cephalosporins	Disruption of the gut microbiota leads to deficiencies in vitamin K due to impaired absorption and bacterial synthesis of the vitamin

Abbreviation: GGCX, γ -glutamyl carboxylase; VKORC1, vitamin K 2,3-epoxide reductase complex; VKA, Vitamin K antagonist.

Because VKCFD is a diagnosis of exclusion, it should be considered when abnormalities persist despite well-controlled vitamin K supplementation at standard doses.²⁹

Prenatal diagnosis of VKCFD is possible with identified familial variants, but not essential, as the condition is treatable with vitamin K. Administering vitamin K during the third trimester can reduce the risk of bleeding in at-risk newborns.

Challenges in Neonatal Diagnosis

Diagnosis of VKCFD in neonates can be challenging due to hepatic immaturity and several factors contributing to idiopathic hypovitaminosis K. In newborns, this condition can progress to neonatal hemorrhagic disease, a serious complication associated with vitamin K deficiency. A detailed evaluation of maternal medication use during pregnancy is critical, as certain medications can interfere with fetal vitamin K metabolism, exacerbating neonatal deficiency. Factors contributing to neonatal hypovitaminosis K include (i) limited fetoplacental transfer of vitamin K, (ii) low levels of vitamin K in breast milk, (iii) an immature gut microbiota unable to synthesize sufficient vitamin K₂, (iv) reduced intestinal absorption, and (v) underdeveloped VKORC1 activity, which is essential for vitamin K recycling. Caution should be exercised when administering antibiotics or other drugs that may interfere with vitamin K metabolism, especially in neonates.^{29,30}

The temporal classification of vitamin K deficiency bleeding (VKDB) helps clinicians diagnose and manage the condition. VKDB is divided into three types: **Early VKDB** (within 24 hours), often linked to maternal use of vitamin K antagonists or other drugs, presenting with severe bleeding like intracranial hemorrhage; **Classical VKDB** (2–7 days), common in exclusively breastfed infants without vitamin K prophylaxis, leading to mucocutaneous or gastrointestinal bleeding; and **Late VKDB** (2 weeks–6 months), associated with high morbidity and mortality, mainly in breastfed infants without supplementation. Prophylaxis at birth is crucial to prevent VKDB.

Treatment

Vitamin K supplementation is the primary treatment for VKCFD, although patient responses can vary significantly. For

individuals with mild to moderate deficiencies, oral vitamin K (e.g., 10 mg two to three times per week) is effective in managing cutaneous and mucosal bleeding and reducing the risk of major hemorrhages. If oral administration is not tolerated, the same dose can be administered intravenously, with dosing intervals adjusted based on PT and international normalized ratio (INR) values.³¹ Despite the known effectiveness of vitamin K, optimal factor level targets for treatment remain unclear, and no fixed therapeutic schedule exists for this rare condition.

The frequency and dosage of vitamin K should be personalized, depending on the patient's symptoms and response. In some cases, coagulation factor activity may not normalize even with high-dose vitamin K, indicating persistent incomplete carboxylation. While protein C (PC) behaves similarly to other vitamin K-dependent proteins, protein S (PS) levels are often only partially restored. Previous studies suggest that the rapid improvement in clotting times is due to the release of uncarboxylated coagulation factors, which undergo γ -carboxylation, while sustained normalization occurs as a result of γ -carboxylation of newly synthesized proteins. Variations in the synthesis pattern may exist for each factor, and alternative reductive pathways, which operate less efficiently under normal conditions, may also help restore the reduced form of vitamin K when supplemented from external sources.

For patients undergoing surgery or experiencing major bleeding, fresh frozen plasma is usually administered at a dose of 15 to 20 mL/kg, with repeated infusions sometimes required to achieve a satisfactory clinical response.⁹ The use of PCC, which contains varying amounts of FII, FVII, FIX, FX, and regulatory proteins C and S, is less well-documented but may be a viable therapeutic alternative, particularly in the treatment of bleeding associated with vitamin K antagonists.³² In addition, recombinant FVIIa (rFVIIa) has been used successfully in patients at low risk of bleeding undergoing surgery.³³

Disabling sequelae are rare but can occur in patients diagnosed after critical events, such as intracranial hemorrhage in neonates. However, most cases are less severe and have favorable outcomes, with no long-term consequences from recurrent bleeding. Widespread vitamin K supplementation at birth and early intervention with appropriate treatment significantly improve prognosis.

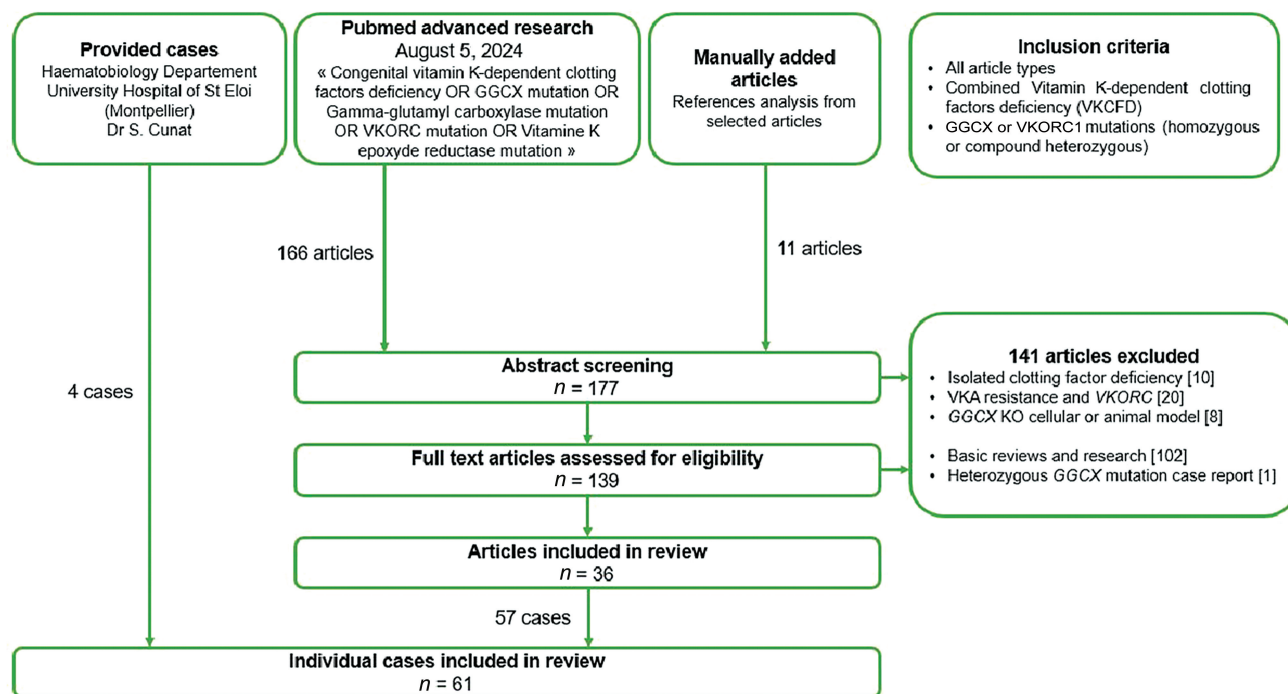


Fig. 1 Overview of article selection and patient data extraction. GGCX, γ -glutamyl carboxylase; VKORC1.

Literature Review of Published Cases and Report of Four New Unpublished Cases from France

The current literature on VKCFD consists mainly of case reports and studies describing variants of *GGCX* and *VKORC1*. Due to the rarity of the disease, meta-analysis is not feasible, and the number of recent reviews is limited.^{9,34} As a result, no study has yet examined all available data or explored the relationship between laboratory findings and clinical outcomes.³⁵ To fill this gap, we conducted a comprehensive review of published cases with *GGCX* or *VKORC1* variants or a typical VKCFD phenotype, as well as several unpublished cases. A systematic search of PubMed identified 166 articles, of which 141 were excluded based on the inclusion criteria. In addition, 11 relevant articles were identified through reference analysis. Finally, 36 articles published between 1966 and 2024 were included according to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) guidelines.³⁶

All types of articles (case reports, case series, reviews, and original studies) were included because of the limited data on VKCFD. A total of 45 cases, covering all age groups, with homozygous or compound heterozygous variants in the *GGCX* (NM_00821.7) or *VKORC1* (NM_024006.6) genes were included from the selected articles (**~Fig. 1**). In addition, four unpublished French cases with identified variants were added. Genetic data were verified using the ClinVar Miner database (September 2, 2024).³⁷ Twelve cases with typical VKCFD phenotypes but without identified variants, due to unavailability of DNA sample or patient refusal, were also included.

In the literature, VKCFD is defined as a deficiency of all four vitamin K-dependent coagulation factors (FII, FVII, FIX, and FX) with levels below 70%. A clinically relevant deficiency is characterized by a reduction of at least one factor below its hemorrhagic threshold, that is, Factor II <20%, Factor VII <20%, Factor IX <40%, or Factor X <30%. Sixty-one patients who met this definition were identified. Their laboratory and genetic characteristics are summarized in **~Table 2**, with additional demographic, genetic, clinical, and laboratory details provided in **~Supplementary Tables S1A to S1C** (available in the online version only). Among the 61 VKCFD patients, 21% were diagnosed in the neonatal period (before 1 month of age), and 34% were diagnosed before reaching 1 year of age. For those diagnosed after the neonatal period, the median age at first evaluation was 10 years (interquartile range [IQR], 1–24). The male-to-female ratio was 0.8 (27 males/34 females). One fetal case was diagnosed following a medical abortion at 19 weeks gestation due to neurological defects.

Relationship between Genetic and Clinical Data Available

Among the 49 patients with identified genetic variants, 82% had *GGCX* variants (38% homozygous), 18% had *VKORC1* variants (all homozygous), and no variants were found in 12 patients (**~Table 3**). The most common *GGCX* defect was in the horizontally transferred transmembrane domain (HTTM), present in 63% of cases. Other defects included the epimerase domain (RmlC-like) in 23%, the vitamin K-dependent protein glutamyl residue-binding site (Glu-BS) in 18%, transmembrane domain 5 (TMD5) in 10%, and the Gla domain in 2 cases. New *GGCX* variants have been identified in unpublished cases: Patient #58 had a compound

Table 2 Patients characteristics of 61 cases reported in this review

Characteristic	All patients	GGCX	VKORC1	No variant
Demographic data, n/N	61/61	40/61	9/61	12/61
Age at first evaluation, n/N	58/61	40/40	6/9	12/12
Neonate ^a , n/N, %	12/58, 21%	8/40, 20%	3/6, 50%	1/12, 8%
Non-neonate, n/N, % (median [IQR])	49/58, 85% (10 [1–24])	32/40, 80% (13 [1.2–31])	3/6, 50% (7 [4.5–7])	6/12, 50% (4 [0.4–19])
< 1 year, n/N, %	20/58, 34%	12/40, 30%	3/6, 50%	5/12, 42%
≥ 1 year, n/N, %	41/58, 71%	28/40, 70%	3/6, 50%	7/12, 58%
Male gender, n/N, % (sex ratio [M/F])	27/61, 44% (0.8 [27/34])	21/40, 53% (0.9 [19/21])	7/9, 78% (0.2 [2/9])	6/12, 50% (1.0 [6/6])
Ethnic groups^b, n/N, %				
European, n/N, %	37/61, 61%	–	–	–
Middle East, n/N, %	13/61, 21%	–	–	–
Genetic data, n/N	49/61			
GGCX defect, n/N, %	40/49, 82%	–	–	–
Homozygous, n/N, %	15/40, 38%	–	–	–
HTTM, n/N, %	25/40, 63%	–	–	–
RmlC-like, n/N, %	9/40, 23%	–	–	–
Glu-BS, n/N, %	7/40, 18%	–	–	–
TMD5, n/N, %	4/40, 10%	–	–	–
Gla domain, n/N, %	2/40, 5%	–	–	–
VKORC1 defect, n/N, %	9/49, 18%	–	–	–
Homozygous, n/N, %	9/9, 100%	–	–	–
ER retention signal defect, n/N, %	9/9, 100%	–	–	–

Abbreviations: ER, endoplasmic reticulum; GGCX, γ-glutamyl carboxylase; Gla domain, autocarboxylation domain; Glu-BS, glutamyl-binding site; HTTM, horizontally transferred transmembrane domain; IQR, interquartile range; PXE, pseudoxanthoma elasticum; RmlC-like, epimerase domain; TMD5, transmembrane domain 5; VKORC1, vitamin K epoxide reductase complex.

^aNeonate: less than 1 month of age.

^bOther ethnic groups: South and Central America, North Africa, and India.

Table 3 Clinical data of 61 cases reported in this review

Characteristic	All patient	GGCX	VKORC1	No data variant
Clinical data	N = 61	N = 40	N = 9	N = 12
No bleeding history, n/N, %	16/61, 26%	12/40, 30%	0/9, 0%	2/12, 17%
Clinically relevant bleeding, n/N, %	45/61, 74%	25/40, 63%	9/9, 100%	11/12, 92%
Provoked bleeding	11/45, 26%	10/25, 42%	0/9, 0%	1/11, 9%
Abnormal bleeding tendency	28/45, 60%	19/25, 76%	9/9, 100%	10/11, 91%
Intracranial bleeding	12/45, 27%	5/25, 20%	2/9, 22%	5/11, 46%
• Intracranial bleeding in <1-year-old	11/12, 92%	5/5, 100%	2/2, 100%	4/5, 80%
• Intracranial bleeding in neonate	5/12, 42%	2/5, 40%	2/2, 100%	1/5, 20%
Non-hemostatic symptoms	29/61, 48%	22/40, 55%	0/9, 0%	7/12, 58%
Keutel-like syndrome	19/61, 31%	15/22, 68%	–	4/7, 57%
• Skeletal disorders	16/19, 84%	12/15, 80%	–	4/4, 100%
• Cardiac disorders	6/19, 32%	6/15, 40%	–	0/4, 0%
PXE-like syndrome	13/61, 8%	10/22, 45%	–	3/12, 25%
• Cutaneous disorders	13/13, 100%	10/10, 100%	–	3/3, 100%
• Ocular disorders	2/13, 15%	2/10, 20%	–	0/3, 0%
• Atherosclerosis	5/13, 38%	3/10, 30%	–	2/3, 67%

Abbreviations: GGCX, γ-glutamyl carboxylase; PXE, pseudoxanthoma elasticum; VKORC1, vitamin K epoxide reductase complex.

Table 4 Relationship between clinical and laboratory data: complete and significant deficiency

Characteristic	All patient	GGCX	VKORC1	No variant
Biological data for complete or incomplete deficiency	N = 37	N = 24	N = 7	N = 6
Complete deficiency^a, n/N, %	29/37, 78%	17/24, 71%	6/7, 86%	6/6, 100%
Abnormal bleeding history	21/29, 72%	10/17, 59%	6/6, 100%	5/6, 83%
Intracranial bleeding	7/29, 24%	2/17, 12%	2/6, 33%	3/6, 50%
Keutel-like syndrome	9/29, 31%	6/17, 35%	0/6, 0%	3/6, 50%
PXE-like syndrome	3/29, 10%	2/17, 12%	0/6, 0%	1/6, 17%
Incomplete deficiency, n/N, %	8/37, 24%	7/24, 33%	1/7, 14%	0/6, 0%
Abnormal bleeding history	5/8, 66%	4/7, 57%	1/1, 100%	–
Intracranial bleeding	0/8, 0%	0/7, 0%	0/1, 0%	–
Keutel-like syndrome	2/8, 25%	2/7, 29%	0/1, 0%	–
PXE-like syndrome	4/8, 50%	4/7, 57%	0/1, 0%	–
Biological data for significant or insignificant deficiency	N = 47	N = 30	N = 8	N = 9
Significant deficiency^b, n/N, %	38/47, 81%	28/30, 93%	3/8, 38%	7/9, 78%
Abnormal bleeding history	29/38, 76%	20/28, 71%	3/3, 100%	6/7, 86%
Intracranial bleeding	9/38, 24%	3/28, 11%	1/3, 33%	5/7, 71%
Keutel-like syndrome	14/38, 37%	10/28, 36%	0/8, 0%	4/7, 57%
PXE-like syndrome	5/38, 13%	5/28, 18%	0/8, 0%	0/7, 0%
Insignificant deficiency, n/N, %	9/47, 19%	2/30, 7%	5/8, 62%	2/9, 22%
Abnormal bleeding history	6/9, 67%	1/2, 50%	5/5, 100%	0/2, 50%
Intracranial bleeding	0/9, 0%	0, 0%	0/5, 0%	0/2, 0%
Keutel-like syndrome	1/9, 11%	1/2, 50%	0/8, 0%	0/2, 0%
PXE-like syndrome	2/9, 22%	1/2, 50%	0/8, 0%	1/2, 50%

Abbreviations: GGCX, γ -glutamyl carboxylase; PXE, pseudoxanthoma elasticum; VKORC1, vitamin K 2,3-epoxide reductase complex.

^aComplete deficiency: deficiency of all four vitamin K-dependent coagulation factors (i.e., FII, FVII, FIX, and FX <70%).

^bSignificant deficiency: deficiency with of at least one vitamin K-dependent coagulation factor below its hemorrhagic threshold (i.e., FII <20%, FVII <20%, FIX <40%, and/or FX <30%).

heterozygous defect with a missense variant (c.182C > T; p. P61L) and a deletion (c.2069_2070del) responsible for defects in the HTTM and Gla domains. Patient #59 had a homozygous missense variant (c.209T > G; p.L70R) responsible for a defect in HTTM. Patient #61 had a compound heterozygous defect with a deletion (c.1165_1167del) and a missense variant (c.826A > T) responsible for defects in HTTM and Glu-BS. In addition, the homozygous variant (c.292C > T) in the *VKORC1* gene observed in patient #60 has been previously reported in VKCFD2 cases.

Among the 61 patients, 74% experienced bleeding. Bleeding was observed in 63% of patients with GGCX variants, 100% of patients with *VKORC1* variants, and 92% of patients with no identified variants. Of the 45 patients who bled, 60% had a high propensity for mucocutaneous bleeding, while 26% had bleeding related to surgery or antibiotic use. Intracranial hemorrhage occurred in 27% of patients, with 92% of these cases occurring before the age of 1 year. Specifically, intracranial hemorrhage occurred in 20% of patients with GGCX variants, 22% of patients with *VKORC1* variants. Joint bleeding was rare, reported in four patients (one trauma-related), and three patients had umbilical cord bleeding during the neonatal period.

Non-hemorrhagic clinical manifestations were observed in 55% of patients with GGCX variants but were not seen in those with *VKORC1* variants. These included Keutel-like syndrome in 31% of patients (84% with skeletal abnormalities and 32% with cardiac issues), PXE-like syndrome in 8% (all with cutaneous manifestations and 15% with ocular abnormalities), and subclinical atherosclerosis in 38% of patients. Cardiac defects were observed in all patients carrying variants in the HTTM domain of GGCX.^{38,39} Interestingly, some patients exhibited mild PXE-like symptoms later in life, suggesting a progressive onset similar to PXE.

The laboratory data are summarized in [Table 4](#). Data were incomplete in 24 cases: 21 because of missing information and 3 because of factor measurements performed after vitamin K supplementation. Among the 37 patients with complete data, clinically relevant VKCFD was associated with more frequent and severe bleeding, especially intracranial bleeding. However, no association was found between factor levels and non-hemorrhagic comorbidities.

Among the four new French patients, none had severe hemorrhagic symptoms. Three were lost to follow-up due to relocation and discontinuation of care, as they did not

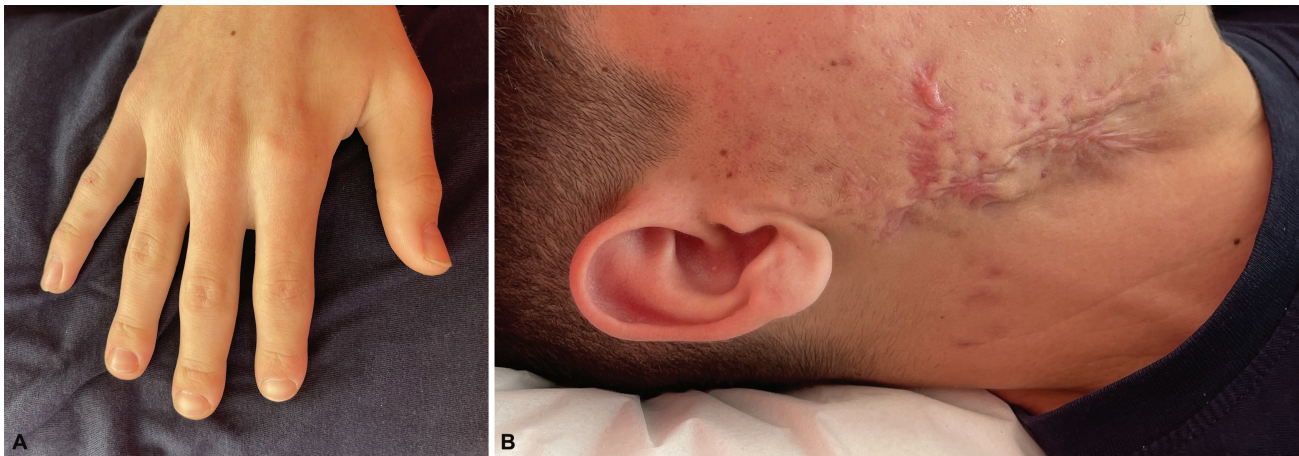


Fig. 2 Clinical signs of a patient with a combined deficiency of vitamin K-dependent coagulation factors: (A) displays shortened distal phalanges in several fingers of the hand and (B) demonstrates the patient's impaired wound healing and cutaneous hyperlaxity.

present with hemorrhagic signs. Only one patient with FII of 23%, FVII of 25%, FIX of 35%, FX of 27%, PC of 19%, and PS of 8%, and a homozygous variant in the *GGCX* gene remains in follow-up. His only bleeding history is a circumcision performed without procoagulant treatment, and he did not have any bleeding complications. He also has short distal phalanges (→Fig. 2A), skin hyperelasticity, and difficulty with wound healing (→Fig. 2B). No cardiologic signs are present.

Conclusions and Future Directions

This review examined all published cases of VKCFD and four unpublished cases with homozygous or heterozygous variants in *GGCX* or *VKORC1*. The cohort exhibited a wide range of clinical presentations, with the severity of bleeding closely related to the degree of vitamin K-dependent factor deficiency. While patients with factor levels below the bleeding risk threshold are at higher risk of bleeding, patients with factor levels above the threshold may still experience bleeding episodes. Moderate bleeding tendencies have been observed more frequently in patients with *VKORC1* variants compared to *GGCX* variants. Homozygous or compound heterozygous variants, especially within the same *GGCX* domain, are often associated with an earlier onset of bleeding, typically before the age of 1 year.³³ Non-hemorrhagic clinical manifestations were more common in patients with *GGCX* variants compared to those with *VKORC1* variants.

Despite these important findings, limitations such as small sample size and missing or non-standardized data remain.

In conclusion, the severity of VKCFD is strongly correlated with the extent of factor deficiency, and compound variants within the same *GGCX* domain lead to more severe clinical disorders. To fill the knowledge gap, the establishment of international registries to collect prospective data on this rare bleeding disorder is essential. Further research is needed to explore the potential role of vitamin K in the defective carboxylation of proteins not involved in coagulation, which could provide deeper insights into the underlying mecha-

nisms of the disease and help identify new therapeutic targets

Authors' Contributions

A.R. conducted the literature review, analyzed the data, and drafted the initial version of the article. S.C. performed the genotyping of the four French patients. C.F., C.O., and A.F. are the physicians responsible for managing the four French patients with VKCFD. Y.D. designed the study, supervised the literature review and data analysis, and finalized the manuscript. All authors reviewed and approved the article's content.

Conflict of Interest

The authors declare that they have no conflict of interest.

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